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Spring toy-inspired soft robots with electrohydraulic actuators

Sohyun Kim[1], Joohyeon Kang[2], Seunghoon Yoo[1] & Youngsu Cha[1]✉

Toys are useful resources for structural inspiration in science and engineering. Their fascinating structures provide new strategies for robotics, particularly in overcoming challenging obstacles and increasing adaptability to unstructured environments. Recent advances in actuators made of soft materials have enabled robots to exhibit safer and more adaptive behaviors during locomotion. However, it is still difficult to descend quickly without falling off at the drop point. In the same context, we recall playing with spring toys descending on stairs. In this paper, we introduce an electrohydraulic-based soft robot inspired by the structure of spring toys. The robot demonstrated a novel and previously unreported ability to descend a series of stairs. Specifically, the soft robot consisted of a helical structure and multiple electrohydraulic actuators. A helical structure was used to accommodate the expansion of the electrohydraulic actuators and to operate a wider range of bending motions. This design prevents unpredictable falls and achieves operation while maintaining a sufficient level of flexibility. We also experimentally investigated the actuation characteristics of the soft robot in terms of motion and force. Additionally, we demonstrated a soft gripper using the spring toy-inspired robot as another potential application.

People can recall playing with toys with curiosity concerning the operating principles. Toys are typically scale models of future realities that have led to interesting scientific applications[1]. Several toys designed for entertainment have served as sources of sophisticated and technological development for engineers from diverse backgrounds[2–4]. For example, the use of kites has led to significant advances in electricity, communication, and flight[5–7]. Toys based on material mixtures have prompted chemical engineers to explore synthetic rubber materials with unique properties[8,9]. The cube display was inspired by the asymmetry of the Rubik's Cube, including its simple structure and intuitive operation[10]. Thus, toys not only have potential engineering applications but also provide some clues to innovative designs that link science and technology.

Among toys with locomotion, we consider spring toys descending the stairs as a representative one. Spring toys can be used as structural inspiration in engineering fields owing to their flexible, helical structures and the locomotion ability to extend to many times their initial length[11–17]. In robotics, a pulse-driven robot capable of self-propulsion has been proposed using a helical structure with a metal loop connected to a pneumatic actuator[16]. The helical structure, which can withstand large deformations, crawls by changing the position of its center of gravity even without limbs. A spring toy-inspired robot that can climb stairs by connecting motors and light springs has been presented[17]. Effectively extended transformations are useful for overcoming challenging obstacles and increasing adaptability in unstructured environments. However, an additional challenge in robotics is to overcome the limitations of structures that can withstand weight and large deformations in terms of stability and versatility. The advent of soft robotics offers an attractive alternative to this problem by possessing compliant bodies and reconfigurable structures[18–20].

The design strategy for soft robotics focuses on exploiting the softness of materials. To achieve flexibility in spring structures, researchers have developed soft robots using origami structures made of soft, thin paper. Folding patterns are designed to achieve the desired elasticity, creating three-dimensional architectures with deployable, reconfigurable, and mechanically unique properties[21–24]. Soft origami robots move on unstructured ground employing rolling propulsion and flipping techniques using thread-motors, pneumatic actuators, and magnetic actuators[25–28]. The bending or folding/unfolding function of origami has been used as a mechanism for grasping objects[29–32]. Although some of them exhibit excellent performance in high-shape deformation and high-speed operations, commonly used operation methods such as electric motors or hydraulic pumps inevitably generate noise and vibration owing to the requirement of bulky, transmission mechanisms. Among various actuation strategies, magnetic actuation is promising for non-contact or remote control of untethered mobile robots[27,28]. However, this is partially limited because it only works in the range of the magnetic field, from a fixed supply device.

[1]School of Electrical Engineering, Korea University, Seoul 02841, Republic of Korea. [2]Department of Smart Convergence, Korea University, Seoul 02841, Republic of Korea. ✉email: ys02@korea.ac.kr

Electrohydraulic actuators have recently been reported as a good method for achieving locomotion, in soft robots, through variable distribution of the internal liquid[33–42]. The actuator consists of a flexible pouch filled with a dielectric fluid and electrodes attached to both sides[33,34]. When a voltage is applied to the electrodes, actuator deformation is caused by induced electrostatic pressure. The actuators do not require either an external source of compressed fluid or a pump for operation. In addition, they exhibit remarkable properties including large operating strain, electromechanical efficiency, fast response, and quiet operation[35–37]. Owing to the favorable characteristics of electrohydraulic actuators, researchers have achieved continuous crawling, jumping, and rolling performance in various environments[38–42]. For example, linear or bending movements generated by modular actuators can be applied to robots that are capable of traversing inclined surfaces[38,39]. The stacked structure of donut-shaped hydraulically amplified self-healing electrostatic, HASEL, actuators achieves vertical jumping via isotropic fluid flow[40]. The anisotropic redistribution structure of the actuator allows it to jump forward quickly and avoid obstacles by generating the energy required for the jump within a very short period[41]. A wheel with an array of multiple and folded actuators rotates and rolls rapidly on the ground. However, it is still difficult, for the designs of actuators developed thus far, to land safely from the drop zone by moving the center of mass[42]. During descent, the actuator may tip over or fall unpredictably owing to structural instability that prevents it from supporting its weight. To solve these challenges, we introduce an electrohydraulic-based soft robot inspired by the structure of spring toys that can move down continuous stairs.

The spring toy-inspired soft robot design utilizes flexible materials and a layered, helical structure that allows for additional connections. Each layer is driven by a soft electrohydraulic actuator. In the proposed design based on this actuation method, bending deformation occurs when the input voltage generates kinetic energy in the helical structure, and then expands the entire structure. Owing to its unique structure, the bending capacity can be fully exploited, and downward movement is realized as the center of gravity of the structure moves. This design approach prevents unpredictable falls owing to its high structural stability and achieves movement on the stairs and slopes while maintaining a sufficient level of flexibility. Soft robots also demonstrate another application with grasping motions by combining two modules.

This work, on spring toy-inspired soft robots, makes three major contributions to the literature. First, the soft robot mimics the elasticity and flexibility of spring structures to achieve a bending deformation through extensive deformation of the body. This means that the tip of the soft robot can be achieved and controlled almost vertically with fewer pieces compared to previous electrohydraulic actuators[40]. Next, the helical structure could stretch several times its original length and descend (moving forward and stopping) a series of stairs instead of rolling to the ground in the drop zone. This is a new feature of descending that has not been previously reported. Finally, the soft robots can also lift up over the half of their own weight as well as perform gently object-grasping motions.

Results

Design and operation principles of spring toy-inspired soft robot

In this study, we propose a spring toy-inspired soft robot with elastic and flexible capabilities driven electrohydraulically (Fig. 1a). The design of the robot has a helical structure, and it is operated using electrohydraulic actuators. A helical structure is a simple origami spring that provides elasticity and can be used to perform a wide range of bending motions when force is applied[12]. The elasticity of the helical structure helps the robot to quickly restore its original shape to avoid being affected by the next deformation. The helical structure was fabricated by coiling layers with a circumscribed circular diameter of 7.5 cm assembled in an octagonal pattern. Paper was used instead of other materials owing to its low cost and flexibility. Additional layers were connected to form a helical structure of the desired length. In addition, electrohydraulic actuators were fabricated consisting of a pouch containing films, fluids, and flexible electrodes for potential wire connections. The pouch was comprised of flexible biaxially oriented polypropylene (BOPP) films and was filled with a dielectric liquid. Carbon conductive tapes were attached on both sides of the pouch surface as a flexible electrode. The wires connected to the electrodes were aligned to the rear of the robot to eliminate the influence of the wires on the motion direction of the soft robot. A prototype of a single electrohydraulic actuator is shown in Fig. 1b. The detailed fabrication process of the helical structure and electrohydraulic actuators has been presented in the Methods.

The fabricated electrohydraulic actuator was operated using electrostatic and hydraulic forces (Fig. 1c)[33,34]. When the input voltage was "OFF," the actuator was resting state. A voltage was applied to the electrodes of the actuator, and charges were accumulated on the electrode surfaces on both sides of the pouch. When the accumulated charge reached a certain threshold, the electrostatic forces attracted the electrodes to each other and squeezed the dielectric liquid. As the zipping process progressed, the cross-section of the non-electrode area was inflated by the fluid being rapidly pushed out. Owing to the inability of the pouch to stretch, the thickness of the cross-section increased, and the length of the sides decreased. This cross-sectional deformation of the electrohydraulic actuators induced smooth bending motions while expanding the helical structure of the robot (Fig. 1d). As can be seen in the conceptual perspective view, two actuators formed a symmetrically placed pair and were attached sequentially along each layer. With this arrangement, the range of bending angles could be customized by adjusting the number of layers and actuators. The attached actuators were divided into two channels and controlled according to the schematic shown in Fig. S2. All actuators within the same channel were connected in parallel and had a common ground. To visualize the operation, schematics consisting of actuators in active and inactive states were partially enlarged and shown in cross-sectional views. The actuators were placed in several layers lying flat in the initial state. When the actuators in one channel were activated, the flat shape inflated and pushed the adjacent stacked actuators to expand the helical structure. This deformation led to an overall bending motion similar to that of pneumatic bending actuators[43–45]. Therefore, the actuation of the electrohydraulic actuators caused the robot body to flip over by promoting the arcuate bending of the entire helical structure.

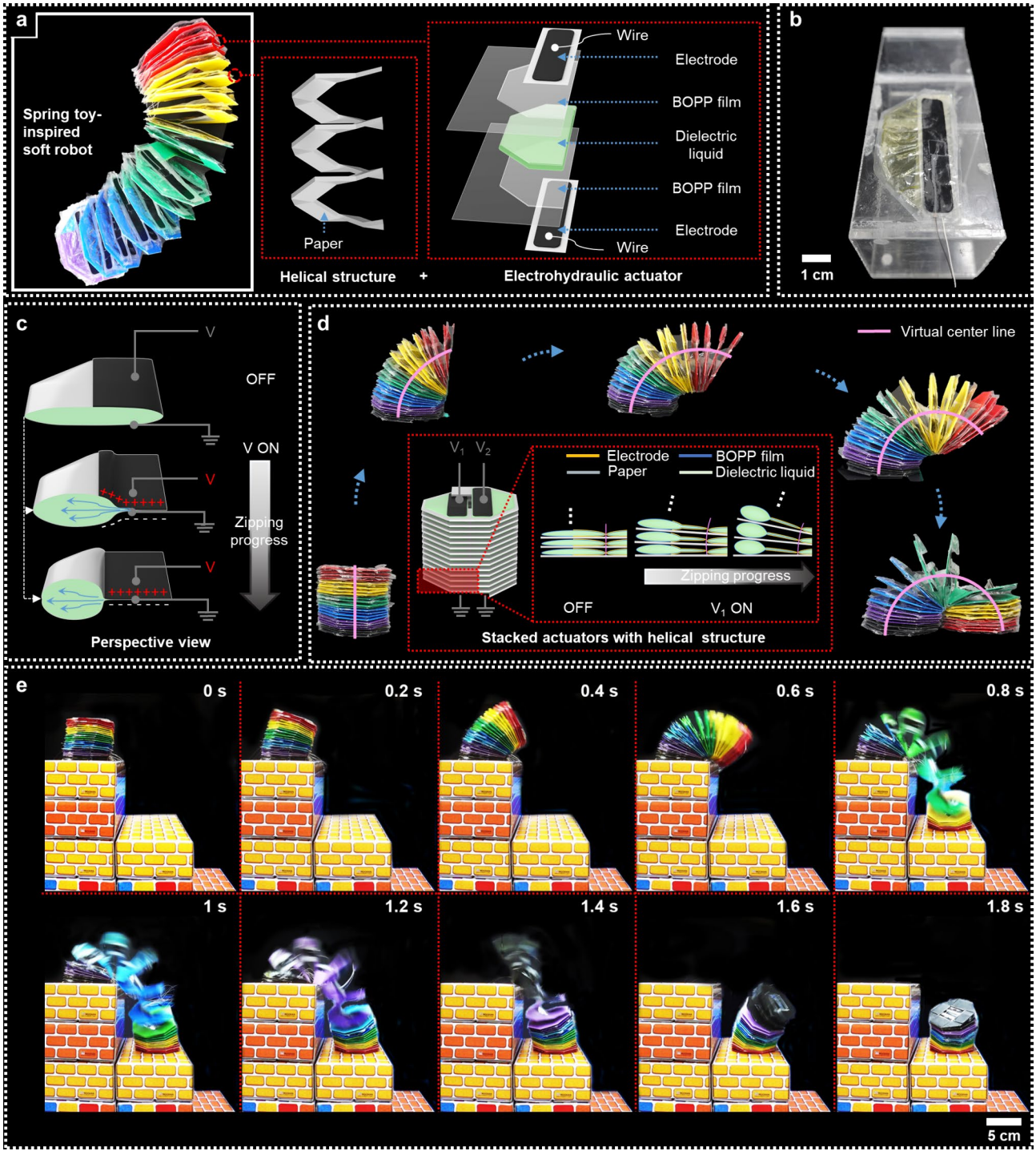


Fig. 1. Design and operation mechanism of spring toy-inspired soft robot. (**a**) Picture and configuration diagram of a soft robot. (**b**) Prototype of the electrohydraulic actuator. (**c**) Operation mechanism of the electrohydraulic actuator at perspective views. By applying high voltage to the two electrodes, the actuator is deformed during the zipping process as the dielectric liquid inside flows to the sides. (**d**) The robot represents the stacked actuators with a helical structure. When voltage is applied to one input channel, an arcuate bending deformation is exhibited along the virtual center line from the deformation of actuators. (**e**) Operating snapshots of a robot descending stairs, when 10 kV is applied to the actuators.

The spring toy-inspired soft robot was stretched to reach the next level, restored to its initial state, and finally walked down the stairs under propulsion. Snapshots of the process of descending the stairs are shown in Fig. 1e. The robot was positioned vertically on top of the stairs. The robot was bent due to the inflation of

the electrohydraulic actuators and was given a propulsion force to move directly onto the stairs below. In other words, the top of the robot flipped over as the helical structure was released towards the bottom. Owing to the elastic properties of its structure, the soft robot descended stably without falling by adjusting the spacing between helices. The robot was restored to its original state during preparation for the next operation. The behavior of the soft robot is shown in Supplementary Video S1.

Bending characterization of spring toy-inspired soft robot

The proposed spring toy-inspired soft robot can induce controllable bending deformations, as shown in Fig. 2a. The bottom of the robot was fixed to the base, and two pairs of actuators were each connected to the two channels for independent control. The bending direction of the robot was determined using the control signal for each channel (top right inset in Fig. 2a). When an input voltage (V_1) is applied to one channel, the connected actuator expands and produces a clockwise bending motion toward the other direction. The bending angle was measured based on the trace markers attached to endpoints M1 and M2 of the soft robot. The bending angle is defined as the variation angle through which the line M1M2 is rotated between the initial and bending states. Here, X is the horizontal axis of the base on which the actuator is placed, and Y is the vertical axis. The detailed experimental setup can be found in the Methods.

The bending angle and blocking force are important measurements that can be used to characterize the bending performance of soft robots (Fig. 2b,c). Under the same circumscribed circle diameter of the soft robot, the number of unit layers and the magnitude of the voltage are important parameters that affect the bending performance. The unit layers are connected sequentially from 10 to 22 floors to represent the various bending angles. In the preliminary tests, unit layers with fewer than 10 floors did not exhibit a noticeable change compared to the initial state angle. Then, an input voltage of up to 10 kV was applied in 2 kV increments from 6 kV to sufficiently reach the compression process of the electrodes. High applied voltages (above ≈ 10.5 kV) were not considered to avoid the electromechanical instability of the actuators[39]. Under the same experimental conditions, five trial repetitions were performed on each actuator sample. The means and standard deviations of the angles were calculated. The average bending angles of the robot reached 19.1°, 26.9°, and 40.8° for the three cases with 10, 14, and 18 floors, respectively, at a voltage of 10 kV. When the layer consisted of 22 floors and a higher input voltage of 10 kV was applied, the average bending angle was larger at 45.9°, whereas the average bending angles at 6 and 8 kV were 24.0° and 39.2°, respectively. In addition, when the number of unit layers was increased from 10 to 18 floors, the average blocking force was increased from 0.27 to 0.48 N at a voltage of 10 kV. The measured average blocking force was the highest approximately 0.54 N for a unit layer of 22 floors, and the corresponding blocking force increased by 245 and 159% for each input voltage of 6 and 8 kV, respectively. Therefore, the bending angles and blocking forces tended to increase as the number of connected unit layers and the magnitude of the input voltage increased. These results demonstrate that significantly improved performance was achieved by adjusting the parameters to the target bending behavior. However, the connection of excessive layers interrupted controllable bending motions and caused the robot body to flip. Accordingly, in this experiment, a layer number of 22 floors or fewer was deemed appropriate for deformation with a controllable bending angle. The movements of each layer of the soft robot are shown in Supplementary Video S2.

Figure 2d shows the bending angles performed while under load by placing a weight on the soft robot. The loads applied to the robot ranged from 3 to 80 g. Herein, the experimental conditions were a unit layer of 22 floors and an input voltage (V_1) of 10 kV based on the results presented above. The robot achieved controllable bending deformations of 41.1°, 29.9°, and 21.0° at loads of 3, 10, and 50 g, respectively. The decrease in bending deformation with increasing external load may be because when a load is applied to a stacked actuator, the dielectric liquid of adjacent actuators is distributed unevenly in the non-electrode area, hindering its operation[35]. In addition to controllable angular output, the robot was strong enough to lift an 80 g load (60% of body weight). The weight was placed vertically on the top of the robot but was observed to collide with the floor due to an imbalance in the center of gravity during the bending operation.

To investigate the operational response of soft robots is important as rapid operation is required for robotic systems to be utilized in applications in industrial environments. Based on the experiments presented above, a unit layer of 22 floors and an input voltage (V_1) of 10 kV were selected for the soft robot operation. The input signal was applied as a square wave for 4 s, after a time delay of 2 s. Figure 3a shows the bending angle of the robot in response to an input voltage signal for 20 s. The inset shows the initial bending angle of the robot in its natural state. In response to the voltage signal, the soft robot bent by approximately 51.2° and then approached a steady state angle of approximately 56.6°. The operation time required to reach 95% of the maximum bending angle at steady state was less than 1 s. As soon as the voltage was released, the soft robot gradually returned to its initial state. This probably means that the charge that accumulates as the voltage is applied continuously requires some time to be released when the voltage is discharged[34,39]. Therefore, the spring toy-inspired soft robot can quickly achieve smooth and continuous bending motion from its natural resting position to the maximum bending angle.

To examine the repeatability of the soft robot, signals of different periods were provided. Based on previous results, applying 10 kV to the robot for at least 1 s was a prerequisite for maximum bending angle. Accordingly, we chose the input frequencies that varied from 0.02 to 0.5 Hz under the same voltage amplitude and pulse duration of 1 s. Figure 3b shows the peak-to-peak bending angle of the robot in a steady state of each logarithmic scaled frequency. Within 0.1 Hz, the bending angle remained approximately constant with a maximum difference of 8% according to the frequency shift. Beyond 0.1 Hz, the angle tended to decrease sharply as the frequency increased. This phenomenon generally may originate from fluid transport[33]. In detail, Fig. 3c and d show the input waveforms at 0.1 and 0.5 Hz and the corresponding bending deformations versus time. In each cycle of frequency experiments, the bending angle was taken within 1 s from the initial to the maximum position. Interestingly, the second peaks were observed at higher frequencies when the voltage was released. This phenomenon may be due

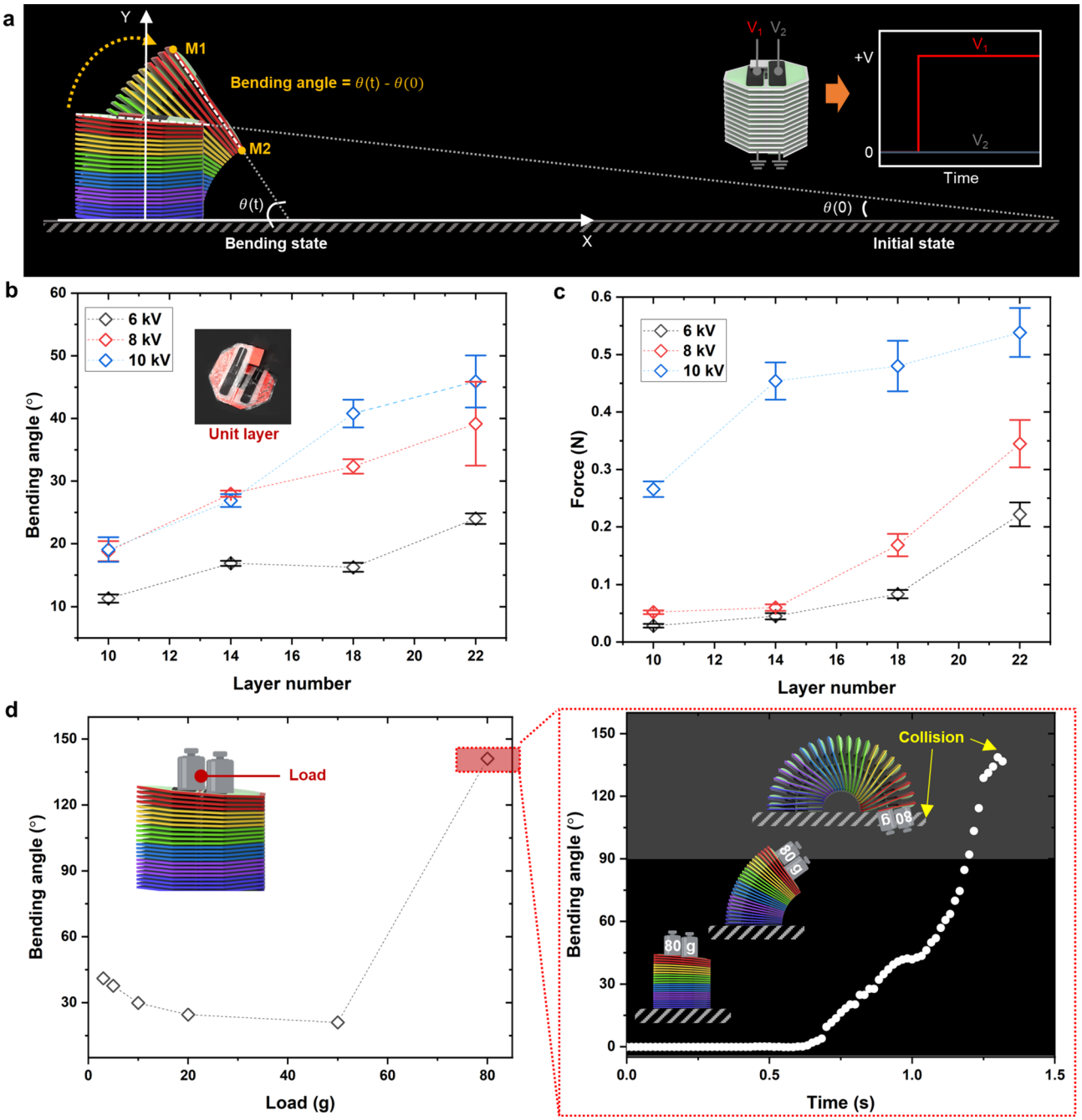


Fig. 2. Bending experiment results. (**a**) Measurement method for bending angle. The bending angle is defined as the variation in angle between the initial and the bending states. The top right inset is the input voltage from the experiment. (**b**) The relationship between bending angle and layer number under different input voltages. The error bars represent the standard deviations for the means for the five experimental trials. (**c**) The relationship between blocking force and layer number under different input voltages. The error bars represent the standard deviations for the means for the five experimental trials. (**d**) The relationship between bending angle and load. Under an applied load of 80 g, the bending actuation of the spring toy-inspired soft robot is stopped due to collision with the floor.

to electrostatic forces related to the square of the AC voltage component of the input signal[34]. We believe that this probably does not allow enough time for the charge to be fully released. Additionally, the robot did not return to its initial position during the recovery cycle for the same reason. In other words, the robot continued to operate in a steady state due to the new angle offset. The results performed well for operating frequencies up to 0.5 Hz.

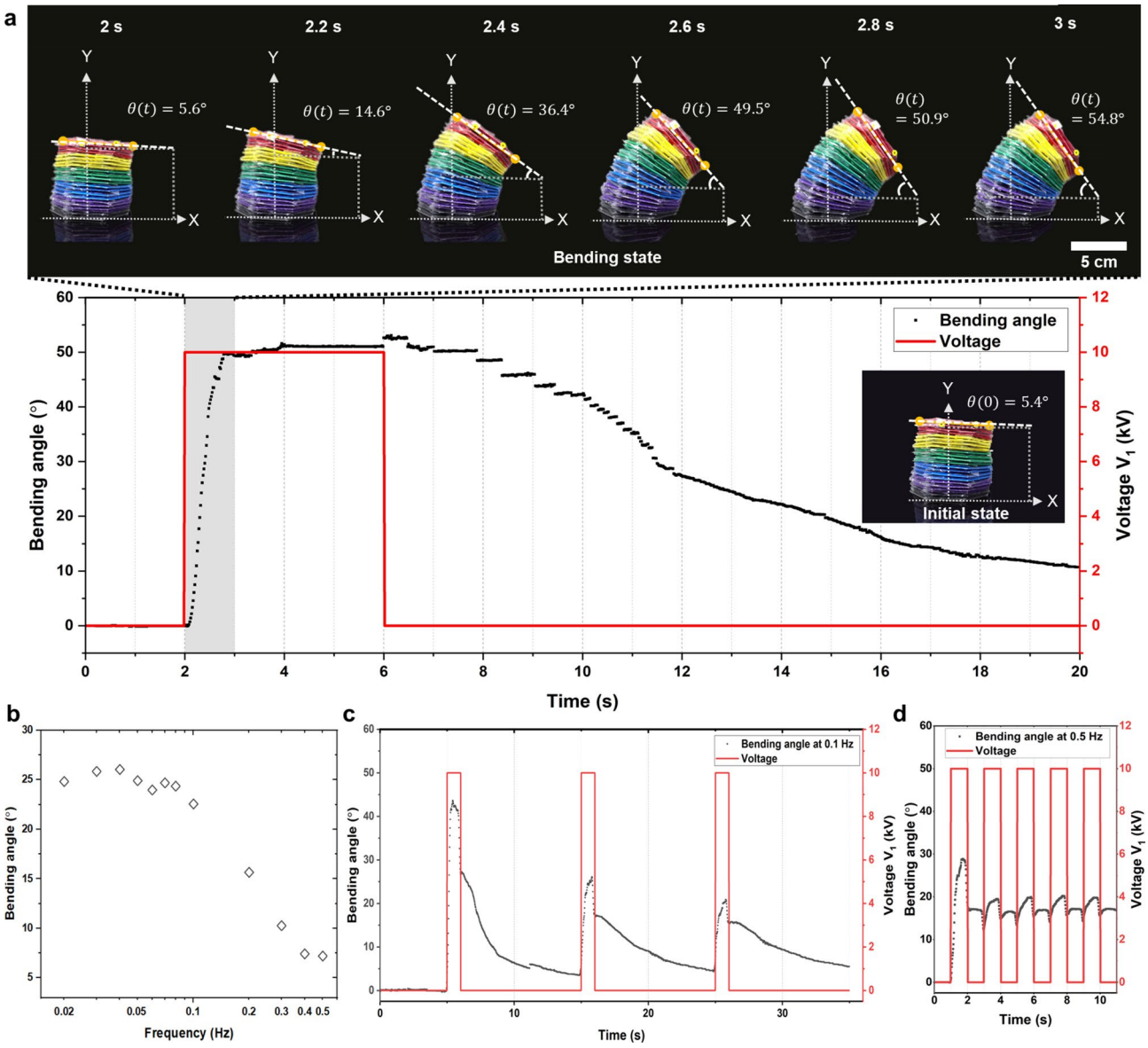


Fig. 3. Continuous bending experiment results. (**a**) Responses of the robot at a square wave of 10 kV when the layer number is 22. The top six snapshot images show the bending behavior of the robot reaching a steady state for 1 s as soon as the voltage is applied. The inset figure represents the initial state. (b) The relationship between bending angle and actuation frequency. Time trajectories from bending angle at (**c**) 0.1 and (**d**) 0.5 Hz.

Expansion characteristics of spring toy-inspired soft robot

The proposed spring toy-inspired soft robot provides a means to generate large actuation strokes by expanding stacked actuators. In particular, when voltage was applied simultaneously to the actuators connected to the two channels, the actuators were inflated equally, causing the body of the soft robot to extend linearly. Using a voltage amplitude of 10 kV and a unit layer of 22 floors based on the experimental results described in Figs. 2 and 3, the robot was operated with expansion in the linear direction (Fig. 4a). To demonstrate the performance of the robot, we measured the Y position of the central point on the top of the robot when the input voltage of a square wave was applied for 4 s. Figure 4b shows the linear strain versus time of the robot over 20 s. The robot achieved up to 52% linear strain within 1 s after the voltage was applied. This response means that the robot can quickly realize good linear expansion motion.

Single stair movements of spring toy-inspired soft robot

Spring toy-inspired soft robots can freely and nimbly descend stairs through extended bending deformations under gravity, showing great potential for future applications. Figure 5a shows the motion sequence of a soft robot descending a single stair, classified into five states. (1) In the initial state, the robot is positioned stably and vertically on the stairs. The soft robot stores gravitational potential energy due to the increased height by being placed at higher points of the stairs. (2) In the bending state, the input voltage (V_1) of one channel was applied

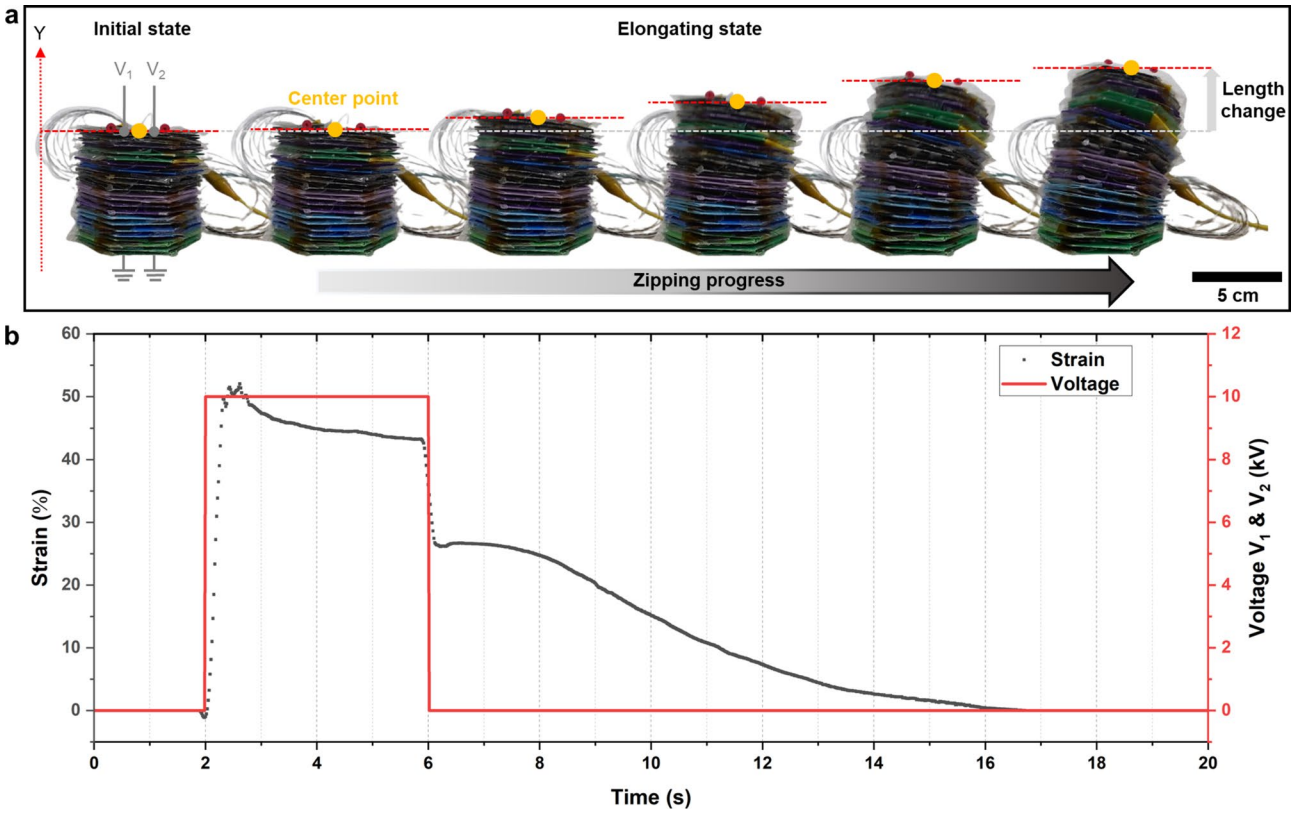


Fig. 4. Linear expansion experiment results. (**a**) Responses of the robot at a square wave of 10 kV when the layer number is 22. The voltage is applied simultaneously to two channels. The snapshot images show the progress of the linear expansion behavior of the robot. (**b**) The center point at the top of the robot reaches 52% strain in the longitudinal direction within 1 s as soon as voltage is applied.

to operate the robot body. The top of the robot was lifted and bent along an imaginary vertical line owing to the expansion of the activated actuators. At this moment, the stored potential energy was converted into kinetic energy and induced a positional shift in the center of gravity. (3) In the next flipping state, the top of the robot unfolded clockwise to the imaginary horizon by gaining momentum under the influence of inertia and gravity. (4) As the top of the robot began to descend into the drop zone, the helical structure expanded and was released downward. (5) Finally, in the restoring state, the soft robot implemented a stable landing motion while the overturned top of the robot was fixed to the base, and the connected helical structure was pulled downward. We demonstrated the movement of the soft robot on a single stair with a height of over 10 cm (Fig. 5b). The number of unit layers connected to the robot for stair movements was 23 floors, and the operating voltage and duration were 10 kV and 1 s, respectively. The movements were recorded at a speed of 960 fps in slow motion mode for detailed observations. During the movements, we defined the start time of the initial state and the end times of the four operational states as 0, T_B, T_{FL}, T_{FA}, and T_R. The processes in each state were captured at equal time intervals, and the total process was presented as 13 snapshots. The robot descended a single stair by bending clockwise and then returned to its original state at the base. Owing to its high speed, the robot completed the task within approximately 2 s. The detailed operating times of the soft robot at various time intervals are presented in Supplementary Table S1.

The specific results were analyzed to obtain insights from the experiments performed in Fig. 5b. The movement of the soft robot was recorded, with trace markers attached at equal intervals along each vertex. Figure 5c shows the trajectories of the experimental coordinates and the theoretical modeling results of the top midpoint of the robot from the bending to flipping states. The robot was modeled as a two-link and two-degree-of-freedom model. According to the Supplementary Materials, the model included a torsion spring in the hinge connecting the massless rigid links, with equal point masses at the hinge and both ends of each link[46]. The equations of motion for the position coordinates of the tip were solved numerically[47]. The experimental and theoretical data were quantified as maximum position coordinates, resulting in arc-like trajectories. The distance error between two trajectories was obtained by dividing the total length by the setting length and aligning them with each other. The differences were then calculated. The root mean square error (RMSE) was 0.1. A good match between the experimental and theoretical results indicated that the mathematical model could accurately predict the bending performance of the robot body. In addition to the trajectory, the curvature and bending angle of the robot are good metrics for quantifying the performance with improved bending. Figure 5d shows the relationship between the curvatures of each row of the robot and time during the procedure from the initial to flipping states. When an input voltage was applied, the entire body of the robot began to bend simultaneously and reached a

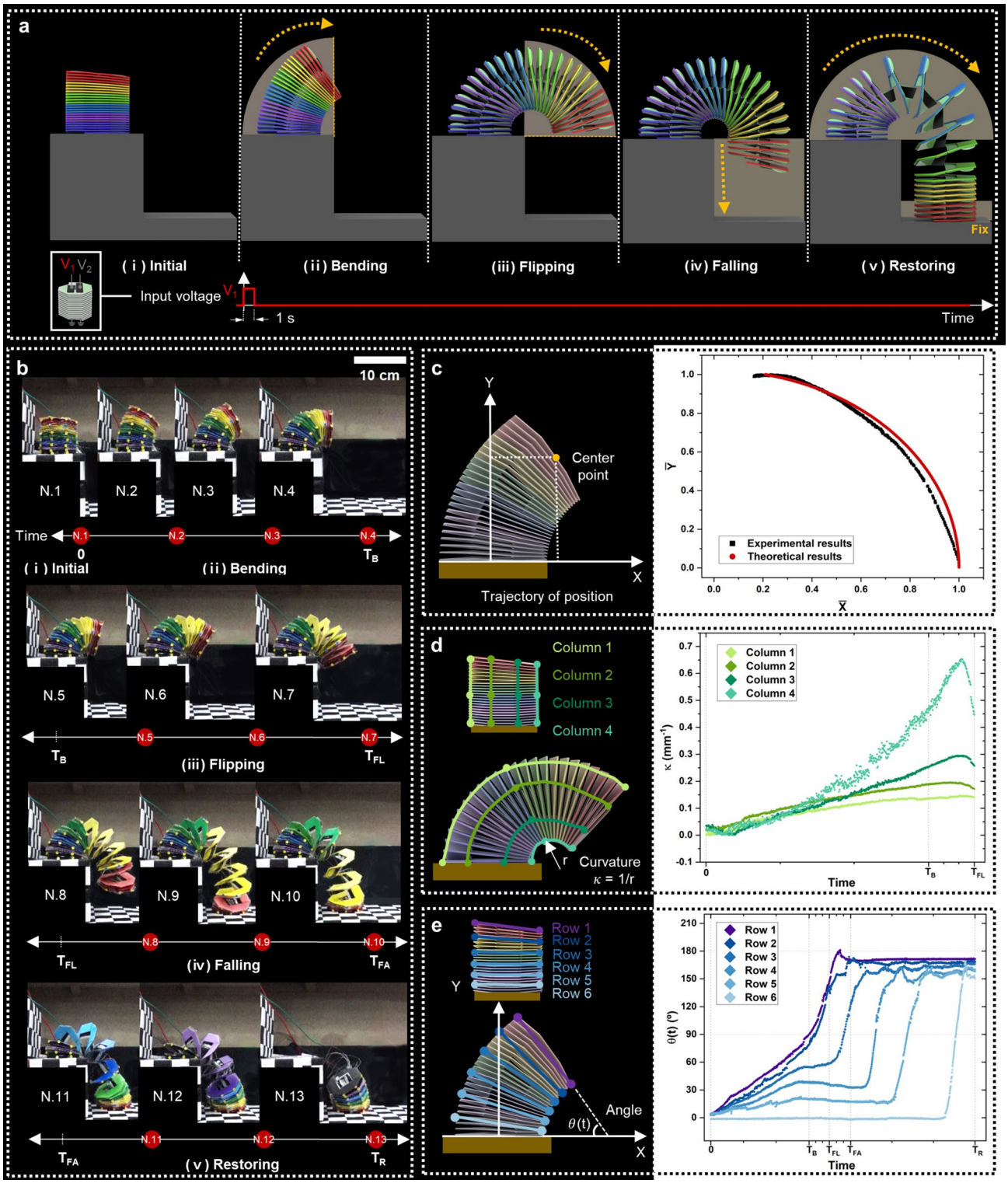


Fig. 5. Moving experiment results on a single stair. (**a**) Schematic diagram of the walking process along a stair of a soft robot with initial, bending, flipping, falling, and restoring states. For operation, an input voltage (V_1) is applied for 1 s to one channel connected to the robot. (**b**) Snapshots of the robot going down the stairs at each state and uniform intervals. The robot takes approximately 2 s to perform a series of tasks. (**c**) Comparison between experimental and theoretical results. The process is investigated from bending to flipping states. (**d**) Time trajectories of curvature of each column. The process is investigated from initial to flipping states. (**e**) Time trajectories of the bending angle of each row. The process is investigated from initial to restoring states.

maximum bending curvature of 0.65 mm^{-1}. In general, the rate of curvature change became faster as the radius of bending became smaller. Figure 5e expresses the angles of each row of the robot from the initial state to the restoring state. The top of the robot gradually increased to approximately 180°, whereas the bottom of the robot represented a sharp transition with a considerably larger slope of the bending change than the other rows. This indicates that the top and bottom of the robot appear different bending speeds owing to the elastic force and gravity from the unique helical structure.

Continuous stair movements of spring toy-inspired soft robot

Previously conducted characterizations and demonstrations showed that the spring toy-inspired soft robot reliably achieved agile stair movements. The continuous stair movements of a soft robot could be realized by adjusting the control signals of the two channels. When the input voltage (V_1) of a square wave represented in Fig. 6a was applied to the actuators connected to one channel, the soft robot walked down the first set of stairs. After stable landing, when an input voltage (V_2) of the same waveform was applied to the actuators connected to the other channel, the robot moved down the second set of stairs using the same mechanism. Therefore, the sequential control signal allowed the robot to perform continuous and controllable downward walking movements. Figure 6b shows the photo sequences of the robot motions, which were flipped up and down from the initial state, and then flipped again and went down by a periodic pulsed signal. The soft robot was proven capable of traversing each stair in less than 2 s, allowing it to adapt better to complex terrains. The continuous motion of the soft robot is presented in Supplementary Video S3.

Slope movements of spring toy-inspired soft robot

To further demonstrate the ability of the spring toy-inspired soft robot to move in various environments, experiments on a sloped surface were performed. The bending angle of a soft robot is affected by the base position due to the direction of the gravity vector[48]. To flip over at a given slope, the soft robot was prepared with a minimum of a unit layer of 14 floors. When the input voltage of the square wave shown in Fig. 7a was applied to the actuators connected to one channel, the soft robot started moving along a flat inclined surface. Figure 7b shows sequential captures of a robot advancing a rubber plate with an inclination angle of 30°. This case differs from Fig. 6 in that the applied signal activates one channel only once during the operating trajectory. That is, the bottom layer began to move continuously at the same time that the top layer completed its flip. The robot moved approximately 40 cm down the slope in 2 s. The continuous slope motion of the soft robot is presented in Supplementary Video S4.

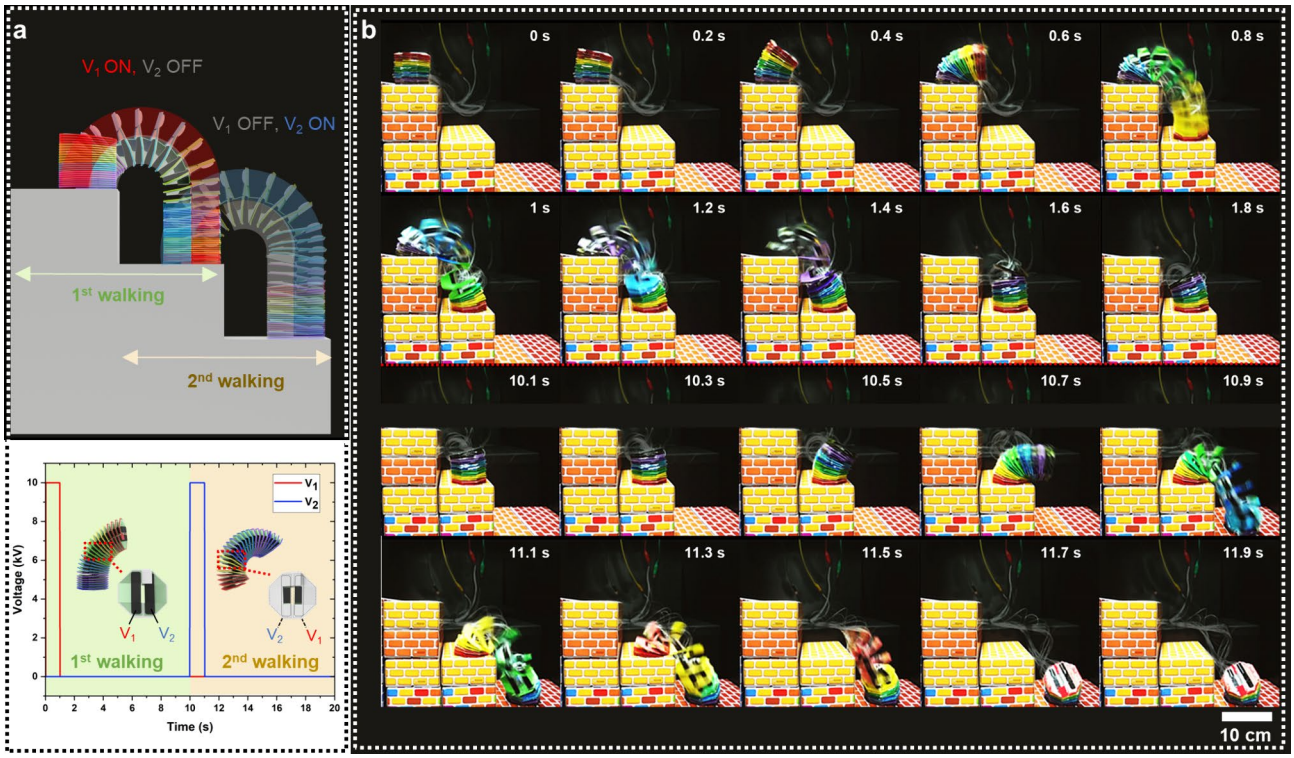


Fig. 6. Moving experiment results on continuous stairs. (**a**) Schematic diagram of the walking process along continuous stairs. Input voltage is sequentially applied to each channel. The inset figures show the enlarged views of the placements of the actuators during each walking process. (**b**) Snapshots of the robot going down the stairs for two steps. The robot takes within 2 s to stably land on the next stair. Control of sequential voltages allows the robot to continuously perform stair movements.

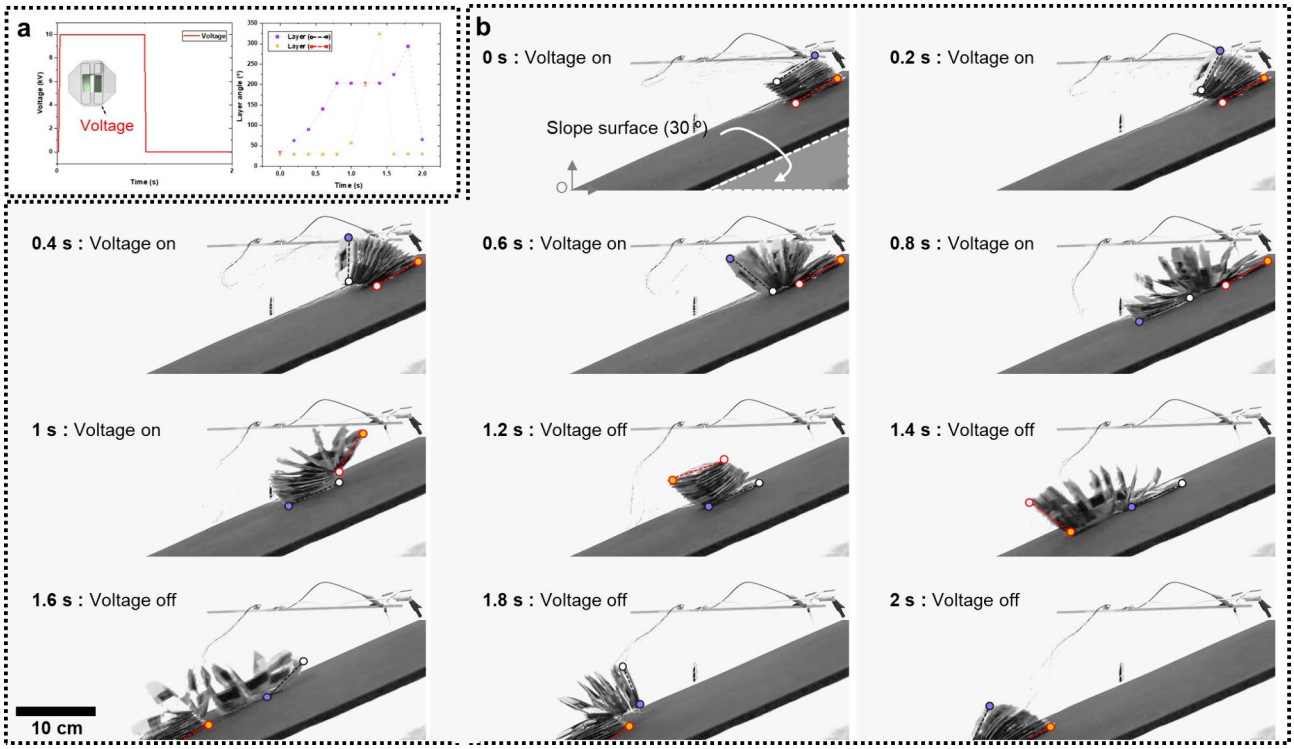


Fig. 7. Moving experiment results on slope. (**a**) Input voltage is applied to one channel. The inset figure shows the location of the electrodes where the voltage will be applied. Right graph shows the time trajectories of the layer angles of the top and bottom. (**b**) Snapshots of the robot moving the slope for two steps. The process is investigated during 2 s.

Application as a soft gripper

Another interesting application of spring toy-inspired soft robots is their use as soft grippers with two modules. The demonstrations were performed using a scaled-down version (1/1.5) of the robot for locomotion. The detailed robot dimensions and experimental methods for the gripper are described in the Methods. The gripper grasped the imitated mangosteen (11.8 g) and plum (8.4 g) objects (Fig. 8a). To understand the delicate manipulation characteristics of the gripper, we investigated the trajectories by tracking the movements of the four tips during the grasping of the plum (Fig. 8b), where T is the period from when the gripper starts moving to when it fully grasps the object. The tips of the gripper approach the target by bending the body by 6/T. Once in contact with the object, it was safely held by modifying the tip positions and producing a gentle touch. An interesting result of the operational performance was that it could bend close to 90° and pick objects of various shapes in approximately 1 s (Fig. 8c). All related videos can be found in Supplementary Videos S5 and S6.

Conclusion

To broaden the applicability of robotics in various fields, continuous efforts are essential to advance the aspects of structural design while using soft and flexible materials. Recent developments in electrohydraulic-based soft mobile robots have demonstrated smooth movements[39] and fast locomotion[41] behaviors over slopes and obstacles, but there remain considerable issues, such as unpredictable falls from drop zones including cliffs. For example, tipping over or slipping in stair movement is another important issue that needs to be handled carefully[49]. The robot must maintain balance and direction of movement to avoid falling down the stairs. From that, the robot body inevitably becomes complex with numerous actuators and sensors to ensure a stable posture. In this study, we proposed an electrohydraulic-based spring toy-inspired soft robot with steerable and fast descent capabilities on stairs and slopes. This design plays an important role in the behavior of mobile robots by increasing stability and enabling fast movement along environments without using external sensors. To the best of our knowledge, this is the first robotic system that utilizes a helical structure for descending walking. The robot maintained a sufficient level of flexibility and accomplished improved bending motions of up to approximately 180°. In addition, the robot stably implemented descending movements by moving the center of gravity of the structure and was restored to its initial state in approximately 2 s on average.

Although this study focused on a robot that walked stairs with a forward descent, it is important to note that the scalability of the structure can be exploited to enhance controllable bending deformations. For example, the maximum bending angle (≈57°) and blocking force (≈0.6 N) can be improved by setting parameters such as the number of unit layers forming the helical structure or the magnitude of the voltage. We also emphasize that our robot can be deformed to angles of up to 180° by connecting more than the critical number of unit layers. This extended bending mechanism allowed the robot to pass over 10 cm of stairs with a short descent

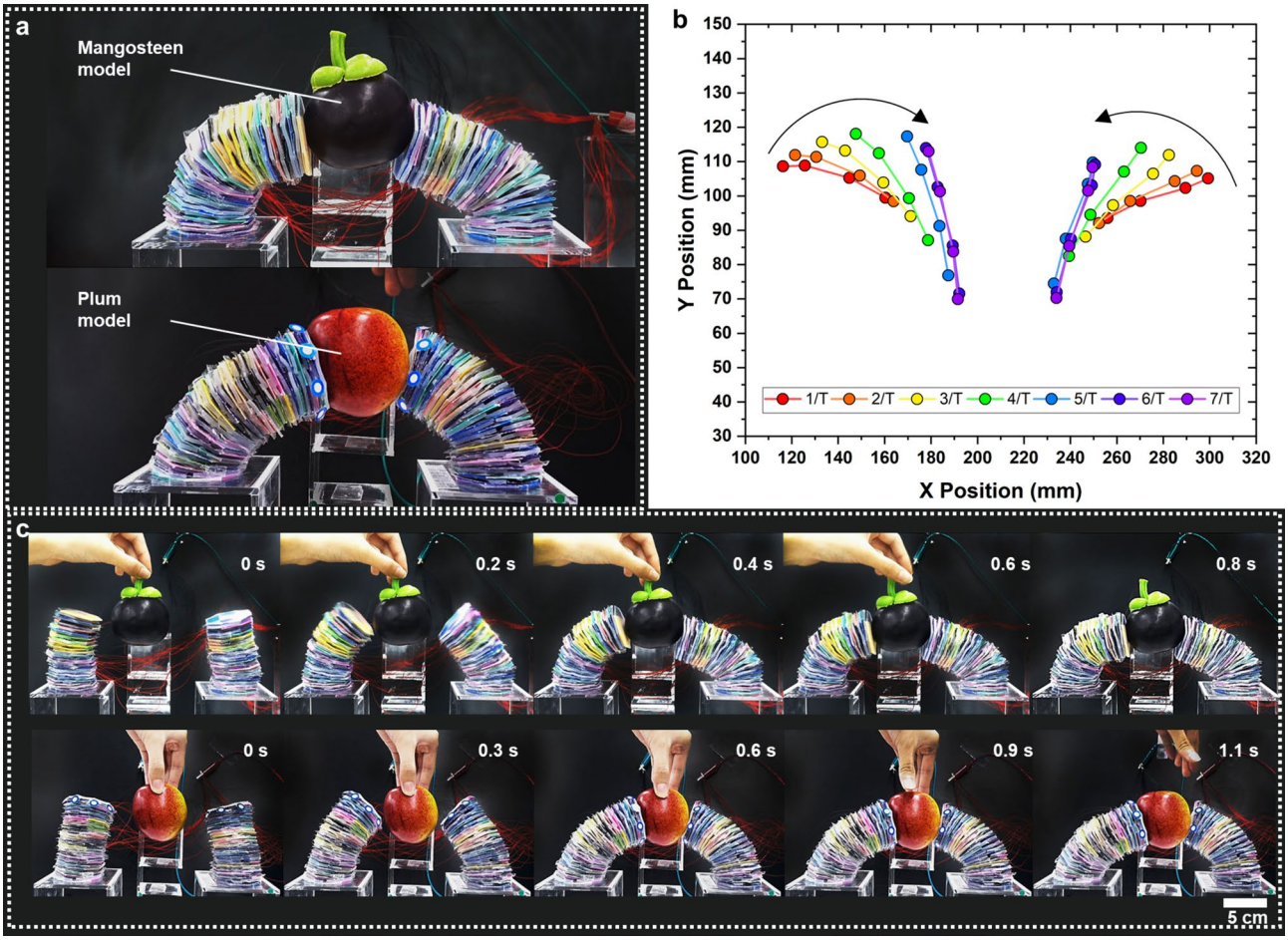


Fig. 8. Application demonstration as a soft gripper. (**a**) The gripper versions of robots grasped the mangosteen and plum. (**b**) The trace markers on the tip are tracked while moving to grasp the plum. The positions of markers are displayed in seven stages from the moment the gripper starts moving until it holds the object stably. (**c**) Snapshots of grasping motions. The gripper takes approximately 1 s to fully grasp each object.

time. Single stair movements were characterized through experimental and theoretical results. A two-link and two-degree-of-freedom mathematical model can predict the targeted trajectories for soft robots and also help explore the dynamics of many other helical structures beyond the prototype fabricated in this study. In addition, the curvatures and bending angles of the robot were investigated in the time domain during descent. Changes in curvatures and bending angles interacting with physical factors can be useful for understanding how robots move downstairs. That is when voltage was applied to the robot, it bent gradually. Subsequently, the robot gained momentum as it went down due to the effects of elasticity, gravity, and inertia, resulting in a fast and stable landing motion. This was explained by the unique helical structure of the robot, which allowed for improved bending and stable descent/landing. Therefore, continuous and controllable downward walking robot movements were achieved through sequential control. Finally, artificial fruit were quickly grasped by two soft robotic modules.

The spring toy-inspired soft robot developed in this work demonstrated fast and large bending deformations and controllability. However, as the number of actuators within the stack increases, the weight of the robot increases or resilience is hampered, ultimately inhibiting the wide range of applications. For example, on horizontal surfaces, the robot has a characteristic of bending in an arch due to the lack of potential energy caused by the height difference. Manufacturing strategies can be adjusted by using other materials, geometries, and fabrication methods to provide larger elasticity and recovery. The actuation performance is significantly affected by the input signal. Continuous input signal with the same polarity causes charge accumulation, which prevents the dielectric liquid from returning to its initial position. Reversing the polarity in successive cycles alleviates the charge retention and achieves fast actuation performance[41]. Future research on this topic could lead to high-performance robots with planar locomotion, bidirectional movement, and orientation-independent manipulation capabilities. To this end, we expect that our approach will improve performance in future works and open previously unknown avenues for realizing next-generation soft robotic systems that provide the necessary resilience.

Methods

Fabrication procedure of soft robot

A new type of spring toy-inspired soft robot with a helical structure and electrohydraulic actuators was developed in this study. The helical structure was an origami spring formed by connecting and stacking unit layers in an octagonal pattern. For the fabrication of the helical structures, colored paper (Jongienara, Korea) with a thickness of 70 μm was selected, as it was inexpensive, flexible, and lightweight. A square piece of paper measuring 7.5 × 7.5 cm was folded into an L-shape. Four such L-shaped pieces were connected sequentially to create a unit layer of the helical structure. An electrohydraulic actuator consists of a pouch containing a dielectric fluid, flexible electrodes, and wiring material. The pouch was prepared by sealing two sheets of BOPP film (Biztem, Korea) with a thickness of 28 μm using a heat press machine (HP3805, Xinhong Mech & Elec, China) using prepared patterns. The heating temperature was 80℃ and the heating time was 40 s. 2 mL of dielectric fluid (FR3 Dielectric Fluid, Cargill, USA) was injected into the pouch. After removing the air bubbles inside the pouch, the liquid inlet was heat-sealed using a heat-bonded machine (Impulse Foot Sealer FI-450/5, Hana Corporation, Korea). Flexible electrodes (Nisshin EM, Japan) were cut into rounded rectangular shapes using a cutting machine (Silhouette Cameo, Silhouette America, USA) to minimize the possibility of electrical damage. Electrodes were attached to both sides of the cut pouch along the pattern edges. Finally, the fabricated actuators formed a pair that was symmetrically placed and sequentially attached along each layer to complete a soft robot with a helical structure (Supplementary Fig. S1).

The gripper version of the soft robots featured a helical structure with unit layers and a circumscribed circle of diameter 5 cm in an octagonal pattern. By adjusting the number of connected unit layers, soft robots can have customizable bending angle ranges. Therefore, the number of unit layers was selected to be 21 floors, which allowed the soft robot to achieve enhanced (> 90°) bending angles to fully pinch and grasp objects of various sizes. In addition, soft robots have thin silicone layers surrounding each layer, which protect the actuators from electrical shorts. To fabricate the silicone layer, a mold with an octagonal pattern of the same diameter as the structure was designed using software (Solidworks, Dassault systems solidworks, USA). The silicone solution was mixed with Ecoflex 0030 Part A (Smooth-On, USA), Ecoflex 0030 Part B (Smooth-On, USA), and a platinum silicone cure accelerator (Smooth-On, USA) at a ratio of 1:1:0.04. The mixture was poured into a mold and cured for 2 h at room temperature.

Circuit configuration of soft robot

Two power amplifiers (MK-200002B, MKPOWER, Korea) and a waveform generator (33500B series, Keysight Technologies, USA) were used for voltage application to the electrodes of each actuator of the soft robot. The output channels of the two power amplifiers were each connected to two pairs of actuators and could be controlled individually. A 100 MΩ resistor was installed in parallel between the power amplifier and actuators for discharge. The input voltage signals for both channels were set using a waveform generator (Supplementary Fig. S2). After the electrical connections were made, the soft robot completed a single stair movement by applying an input voltage of 10 kV to one channel. The voltage waveform was a square wave with an applied time of 1 s. In the continuous stair movement test, voltage signals of the same magnitude and waveform were applied sequentially to both channels with a 10 s interval. Actuation cycles were set at a frequency of 0.5 Hz to ensure sufficient operation time for the soft robot.

As with typical capacitive actuators, the maximum applied electric field in soft robots is limited by the dielectric strength of the materials of construction. From the BOPP film material used, the theoretical breakdown voltage of the actuator was calculated to be 39.2 kV (using 700 V μm^{-1})[50]. However, actuators may not avoid electrical damage due to leakage of the dielectric liquid from small holes created through the heat seal or electrical discharge through the air. For stable operation without dielectric breakdown, the breakdown voltage was much lower (~ 50%)[50].

Measurement setup of soft robot

To observe the deformation of the spring toy-inspired soft robot, trace markers were attached to the sides of the helical structure during the experiments. To obtain the bending angle and position coordinates of the top of the robot, markers were placed at both ends and center. In addition, the markers were evenly positioned along the length of the robot to measure the curvatures of the four columns and bending angles of the six rows. In the application test, the positions of the markers on the four tips of the top were monitored. A camera (DSC-RX100M4, Sony, Japan) was used to capture the dynamic deformation motion. The camera was placed in front of the robot, and most cases were recorded at 60 fps. Experiments on a single stair (Fig. 2b–e) were captured at 960 fps for detailed observation in the slow-motion mode. Optical data, including coordinates and angles, were acquired using a motion tracking program (ProAnalyst Motion Analysis Software, Xcitex, USA). In addition, a load cell (GS0-500, Transducer Techniques, USA) was installed to measure the blocking force of the robot. When the actuators were activated, the load cell was positioned perpendicular to the edge of the inflated parts. The outputs measured by the load cell were amplified using an amplifier unit (IL-1000 amplifier unit, Keyence Co., Japan). Data were collected at a sampling frequency of 2000 Hz using a data acquisition device (NI 9229, National Instruments, USA) (Supplementary Fig. S3). In the grasping tests, the gripper was tested with two modules fixed to the base at 15 cm intervals.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions
S.K. and Y.C. proposed the study. S.K. and J.K. fabricated actuator. S.K., J.K., S.Y., and Y.C. performed methodology. S.K. and S.Y. analyzed the data and wrote original draft. S.K. and Y.C. participated in completing the manuscript.

Competing interests
The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to Y.C.

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